

Chapter 2

Getting to Know Your Data

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Contents

- Data objects and attribute types
- Basic statistical descriptions of data
- Data visualization
- Measuring data similarity and dissimilarity

Data Objects

- Data sets are made up of data objects
- A **data object** represents an entity
- Also called *samples* , *examples*, *instances*, *data points*, *objects*, *tuples*
- Data objects are described by **attributes**
- Database rows -> data objects; columns -> attributes

The diagram illustrates the relationship between data objects and database structure. A table is shown with five columns and four rows. The columns are labeled 'age', 'income', 'student', 'credit_rating', and 'buys_computer'. The rows represent individual data objects. A red bracket on the left side of the table, labeled 'samples', groups the rows. A red bracket above the table, labeled 'attributes', groups the columns.

age	income	student	credit_rating	buys_computer
<=30	high	no	fair	no
<=30	high	no	excellent	no
31...40	high	no	fair	yes

Attributes

- **Attribute** (or *dimensions, features, variables*):
a data field, representing a characteristic or feature of a data object.
 - *E.g., customer_ID, name, address, gender*
- Types:
 - Nominal
 - Binary
 - Ordinal
 - Numeric

Attribute Types

- **Nominal**: categories, states, or “names of things”
 - *Hair_color* = {*auburn*, *black*, *blond*, *brown*, *grey*, *red*, *white*}
 - marital status, occupation, ID numbers, zip codes
- **Binary**
 - Nominal attribute with only 2 states (0 and 1)
 - Symmetric binary: both outcomes equally important
 - e.g., gender
 - Asymmetric binary: outcomes not equally important.
 - e.g., medical test (positive vs. negative)
 - Convention: assign 1 to most important outcome (e.g., HIV positive)

Attribute Types

- Ordinal

- Values have a meaningful order (ranking) but magnitude between successive values is not known.
- *Opinion*= {*strongly disagree*, *disagree*, *neutral*, *agree*, *strongly agree*}, grades, army rankings

- Numeric

- Measured on a scale of equal-sized units
- Values have order
 - E.g. temperature, height, weight, age, and money

Discrete vs. Continuous Attributes

- **Discrete Attribute**
 - Has only a finite or countably infinite set of values
 - E.g. zip codes, profession, or the set of words in a collection of documents
 - Sometimes, represented as integer variables
 - Note: Binary attributes are a special case of discrete attributes
- **Continuous Attribute**
 - Has real numbers as attribute values
 - E.g., temperature, height, or weight
 - Practically, real values can only be measured and represented using a finite number of digits
 - Continuous attributes are typically represented as floating-point variables

Basic Statistical Descriptions of Data

- Measuring the central tendency
- Measuring the dispersion of data
- Graphic displays of basic statistical descriptions of data

Measuring the Central Tendency

- Mean

$$\bar{x} = \frac{1}{n} \sum_{i=1}^n x_i$$

$$\bar{x} = \frac{\sum_{i=1}^n w_i x_i}{\sum_{i=1}^n w_i}$$

- Weighted arithmetic mean:
- Trimmed mean: chopping extreme values

- Median

- middle value if odd number of values, or average of the middle two values otherwise
- suitable for skewed data

Measuring the Central Tendency

- Mode
 - value that occurs most frequently in the data
 - unimodal, bimodal, trimodal
 - if each data value occurs only once, then there is no mode
- Midrange
 - the average of the largest and smallest values in the data

Measuring the Central Tendency

data type	measurement
numeric	mean, mode, median
ordinal	mode, median
nominal	mode

Measuring the Central Tendency

Example: the values for salary (in thousands of dollars)
in increasing order: 30, 36, 47, 50, 52, 52, 56, 60, 63, 70, 70, 110

$$\bar{x} = \frac{30 + 36 + 47 + 50 + 52 + 52 + 56 + 60 + 63 + 70 + 70 + 110}{12}$$

$$= \frac{696}{12} = 58 = \$58,000$$

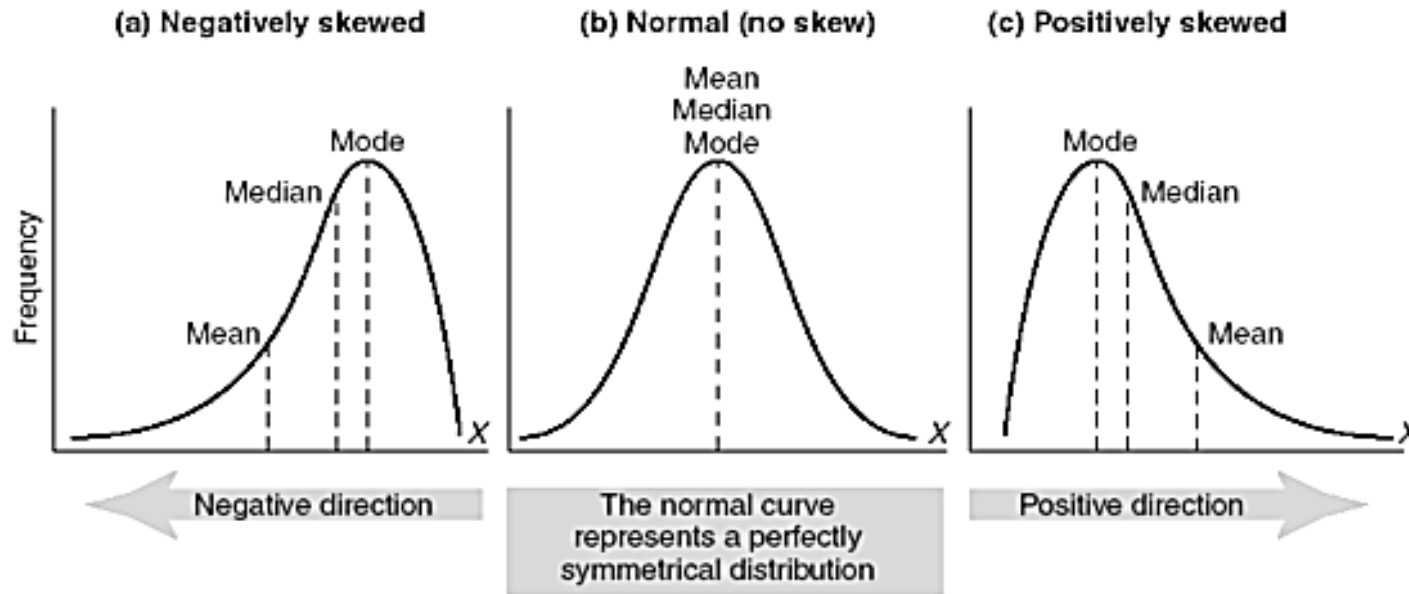
$$\text{median} = \frac{52 + 56}{2} = \frac{108}{2} = 54 = \$54,000$$

bimodal: \$52,000 and \$70,000

$$\text{midrange} = \frac{30,000 + 110,000}{2} = \$70,000$$

Symmetric vs. Skewed Data

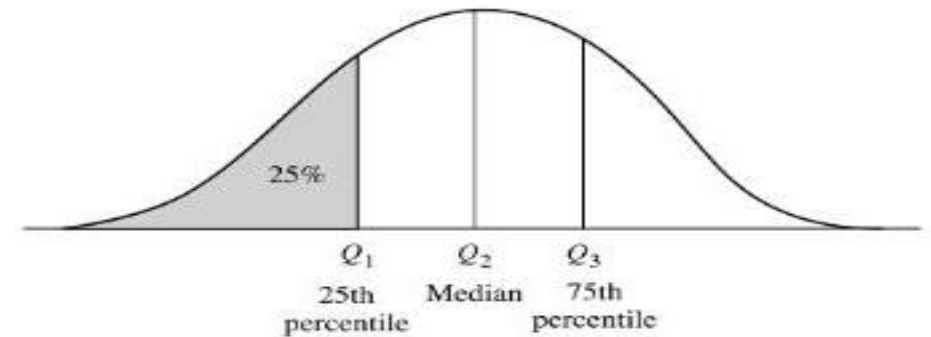
- median, mean and mode of symmetric, positively and negatively skewed data



Measuring the Dispersion of Data

- Quartiles: Q_1 (25th percentile), Q_3 (75th percentile)
- Inter-quartile Range (IQR)

$$IQR = Q_3 - Q_1$$



Example: 30, 36, 47, 50, 52, 52, 56, 60, 63, 70, 70, 110

using Matlab

$$Q_1 = 48.5 \quad Q_2 = 54 \quad Q_3 = 66.5$$

$$IQR = 66.5 - 48.5 = 18$$

Measuring the Dispersion of Data

- Five number summary: min, Q_1 , median, Q_3 , max
- Outlier
 - x is extreme low if $x < Q_1 - (1.5 \times \text{IQR})$
 - x is extreme high if $x > Q_3 + (1.5 \times \text{IQR})$
- Variance: (sample: s^2 , population: σ^2)

$$s^2 = \frac{1}{n-1} \sum_{i=1}^n (x_i - \bar{x})^2 \qquad \sigma^2 = \frac{1}{N} \sum_{i=1}^n (x_i - \mu)^2$$

- Standard deviation s (or σ) is the square root of variance

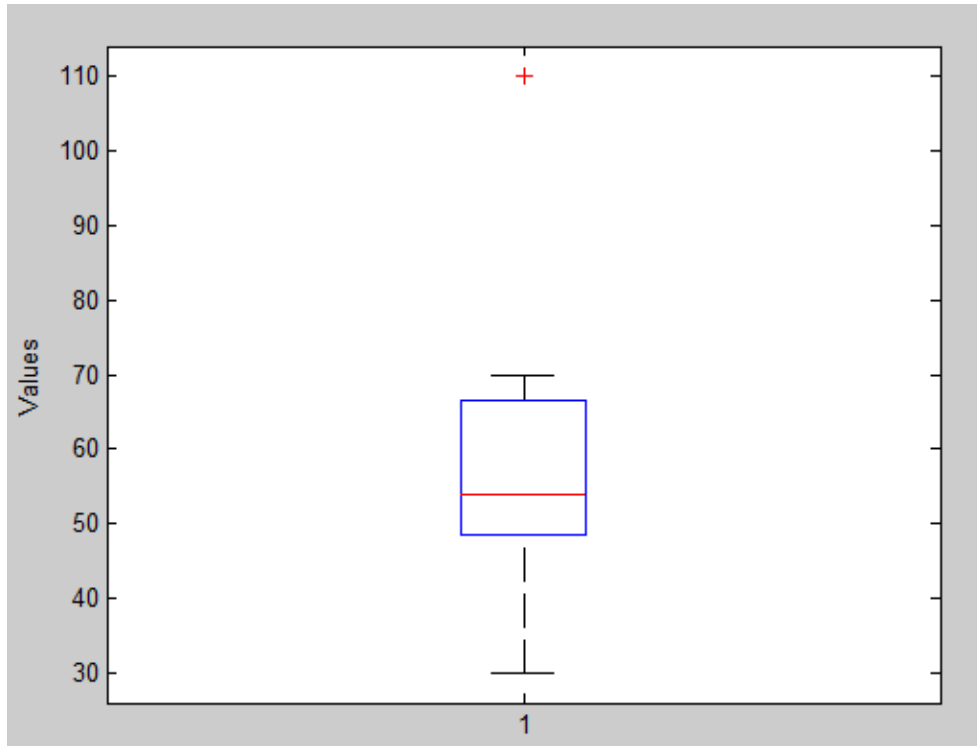
Graphic Displays of Basic statistical Descriptions

- **Boxplot**

- Data is represented with a box
- The ends of the box are at the first and third quartiles, i.e., the height of the box is IQR
- The median is marked by a line within the box
- Whiskers: two lines outside the box extended to Minimum and Maximum
- Outliers: points beyond a specified outlier threshold, plotted individually

Boxplot

Example: 30, 36, 47, 50, 52, 52, 56, 60, 63, 70, 70, 110



$$Q_1 = 48.5$$

$$Q_2 = 54$$

$$Q_3 = 66.5$$

$$IQR = 18$$

Outlier

- x is extreme low if $x < Q_1 - (1.5 \times IQR)$
- x is extreme high if $x > Q_3 + (1.5 \times IQR)$

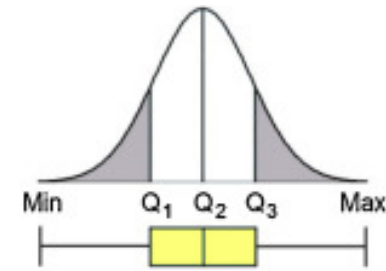
$$\begin{aligned} x &< 48.5 - (1.5 \times 18) \\ &< 48.5 - 27 \\ &< 21.5 \end{aligned}$$

$$\begin{aligned} x &> 66.5 + (1.5 \times 18) \\ &> 66.5 + 27 \\ &> 93.5 \end{aligned}$$

Boxplot

Normal Distribution

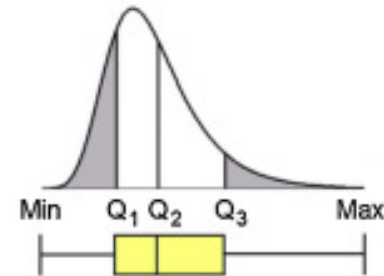
$$(\text{Quartile 3} - \text{Quartile 2}) = (\text{Quartile 2} - \text{Quartile 1})$$



Symmetric

Positive Skew

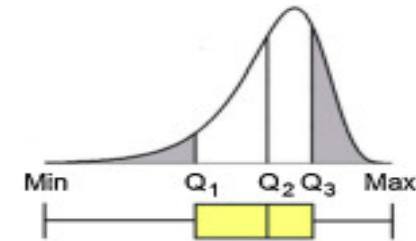
$$(\text{Quartile 3} - \text{Quartile 2}) > (\text{Quartile 2} - \text{Quartile 1})$$



Asymmetric
(positive or right skewed)

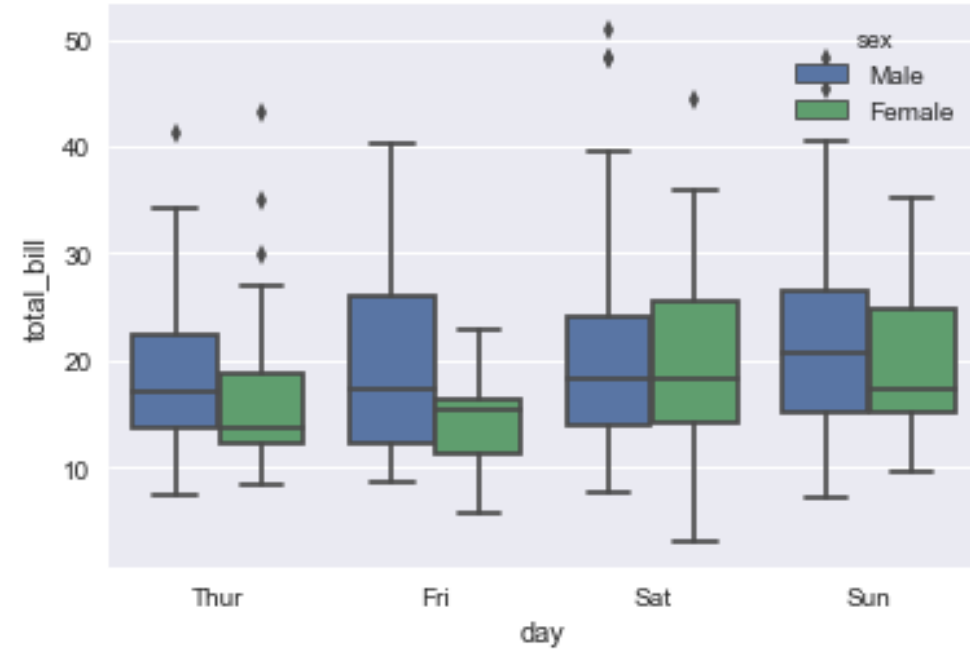
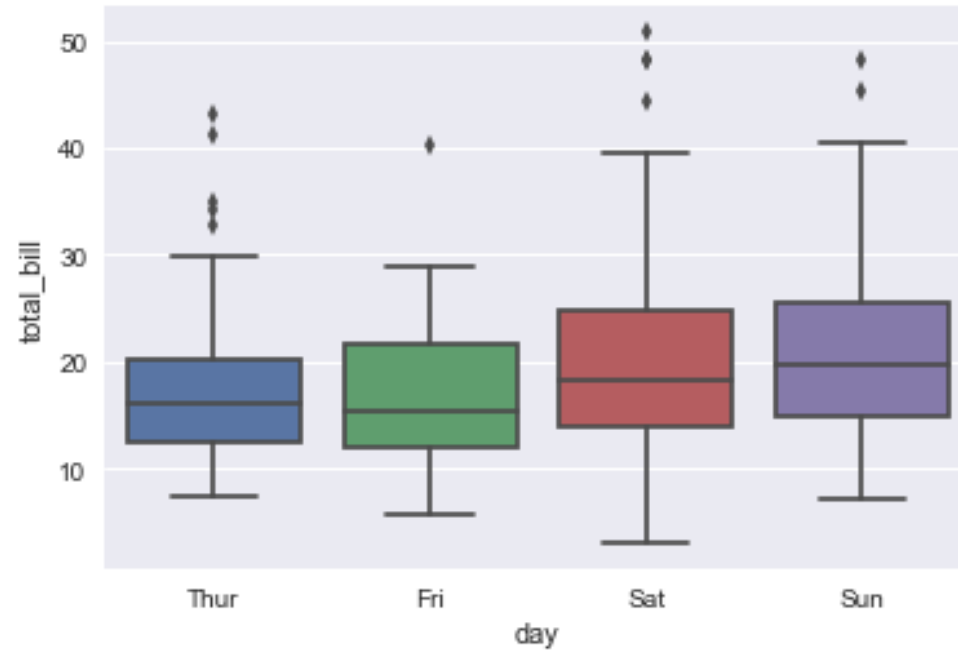
Negative Skew

$$(\text{Quartile 3} - \text{Quartile 2}) < (\text{Quartile 2} - \text{Quartile 1})$$



Asymmetric
(negative or left skewed)

Boxplot



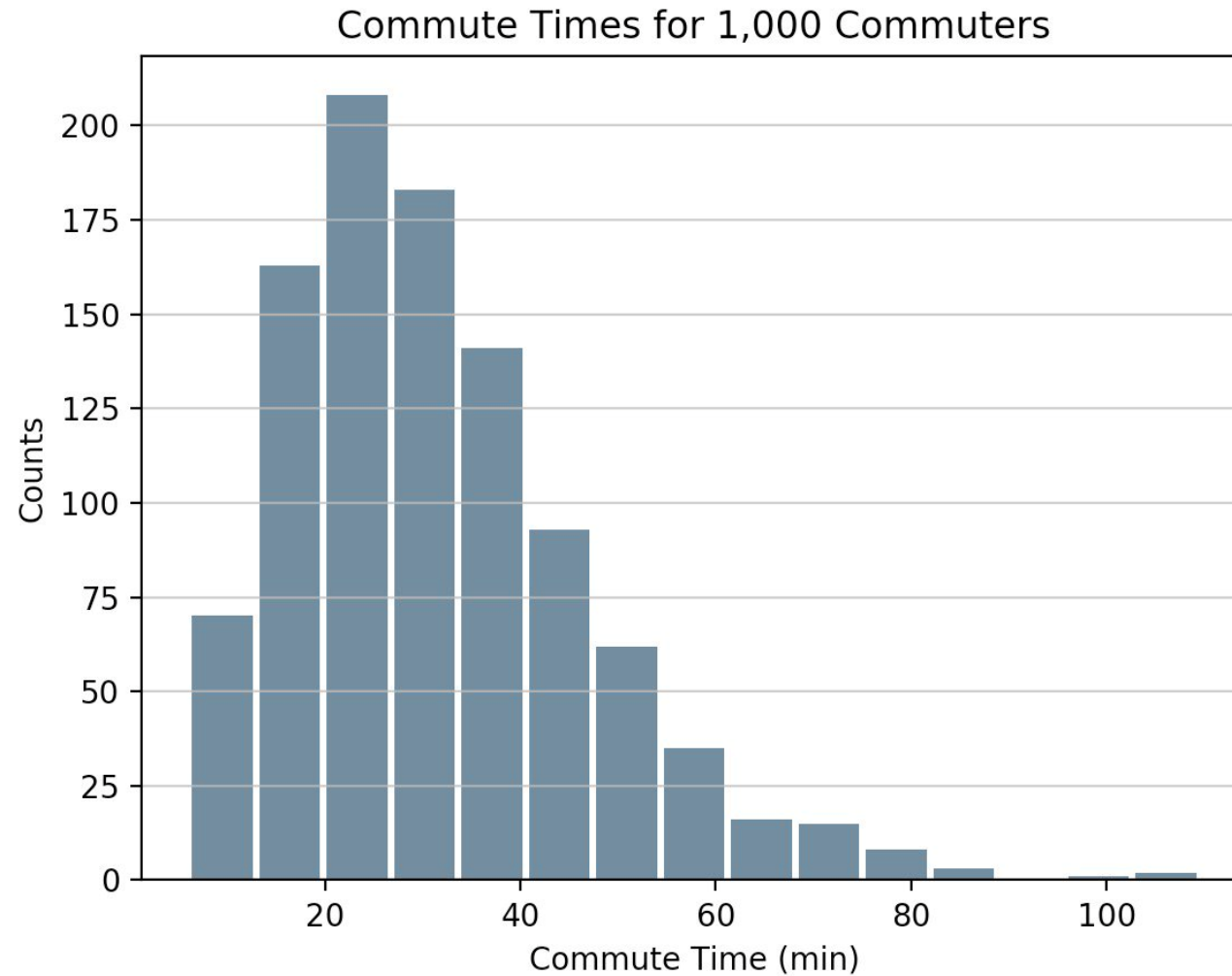
Source: <http://www.learningaboutelectronics.com/Articles/How-to-create-a-box-plot-in-seaborn-with-Python.php>

Graphic Displays of Basic statistical Descriptions

- **Histogram**

- Graph display of tabulated frequencies, shown as bars
- It shows what proportion of cases fall into each of several categories
- Differs from a bar chart in that it is the *area* of the bar that denotes the value, not the height as in bar charts
- The categories are usually specified as non-overlapping intervals of some variable

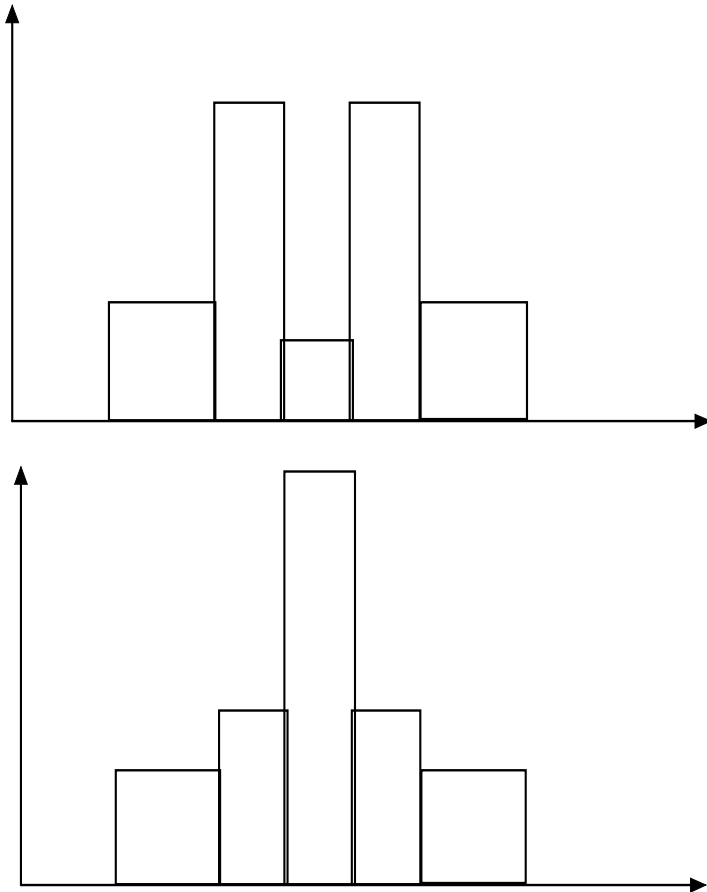
Histogram



Source: <https://realpython.com/python-histograms/>

Histogram

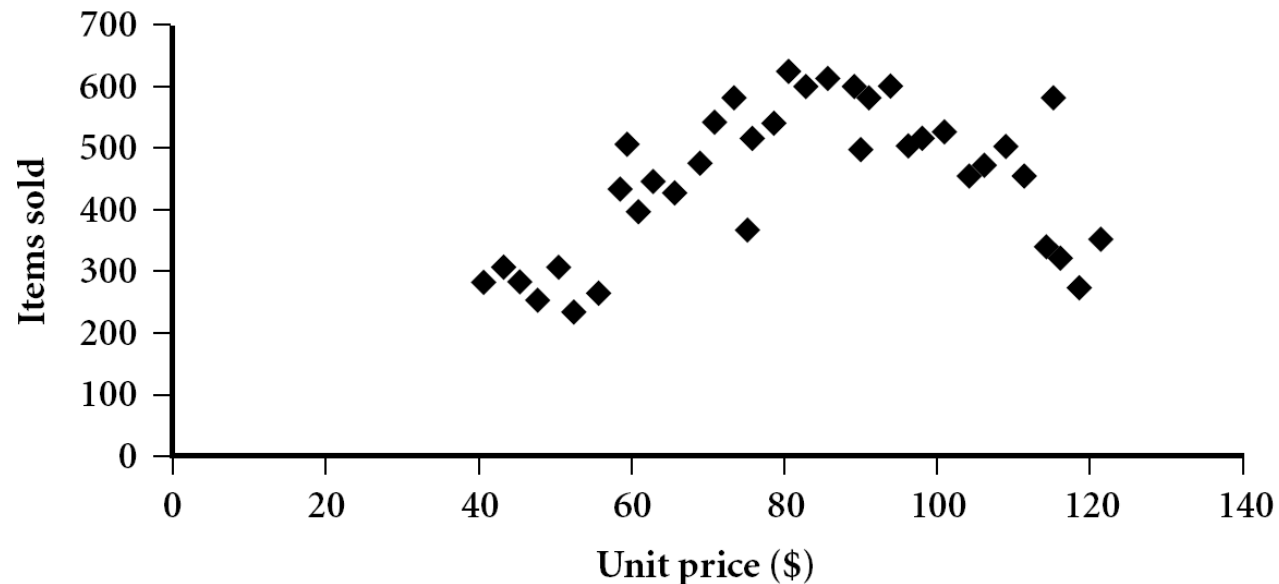
- Histograms Often Tell More than Boxplots



- The two histograms shown in the left may have the same boxplot representation
 - The same values for: min, Q1, median, Q3, max
- But they have rather different data distributions

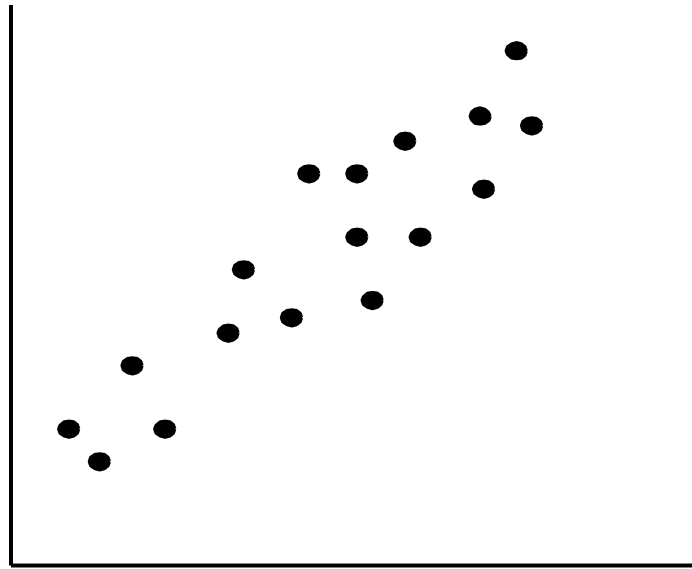
Graphic Displays of Basic statistical Descriptions

- **Scatter plot**
 - provides a first look at bivariate data to see clusters of points, outlier
 - each pair of values is treated as a pair of coordinates and plotted as points in the plane

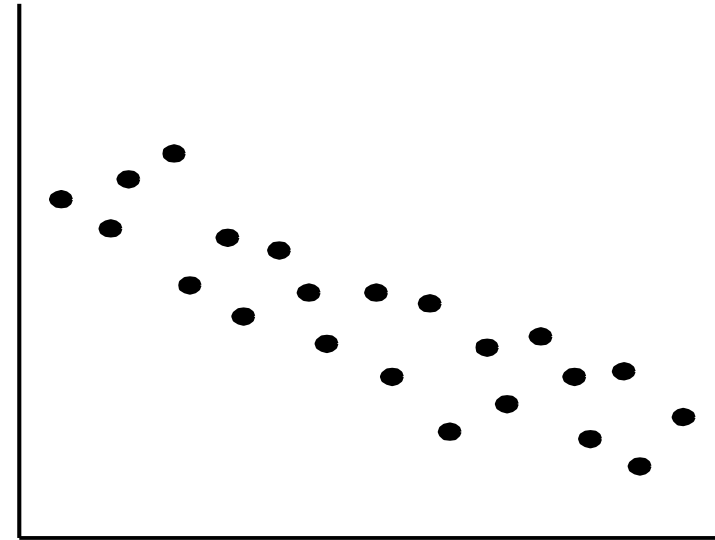


Graphic Displays of Basic statistical Descriptions

- positively and negatively correlated data



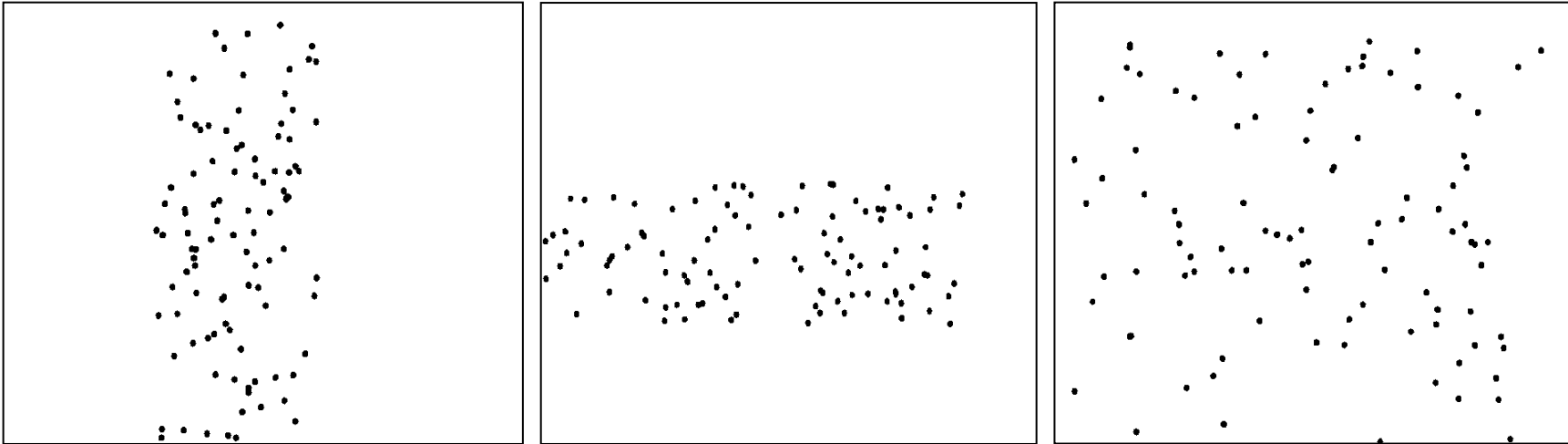
positively correlated



negatively correlated

Graphic Displays of Basic statistical Descriptions

- uncorrelated data



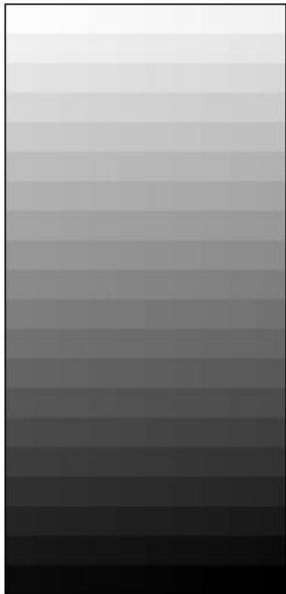
Data Visualization

- Why data visualization?
 - provide qualitative overview of large data sets
 - search for patterns, trends, structure, irregularities, relationships among data
 - help find interesting regions and suitable parameters for further quantitative analysis
- Visualization techniques
 - Pixel-Oriented
 - Geometric Projection
 - Sankey Diagram
 - Icon-Based

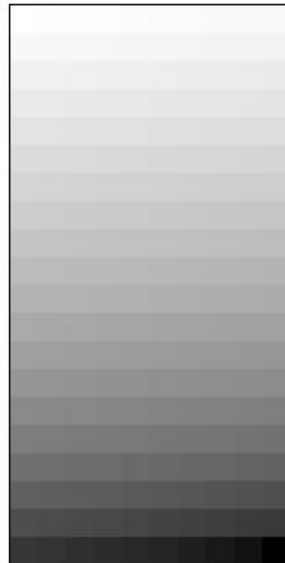
Pixel-Oriented

- For a data set of m dimensions, create m windows on the screen, one for each dimension
- The m dimension values of a record are mapped to m pixels at the corresponding positions in the windows
- The colors of the pixels reflect the corresponding values

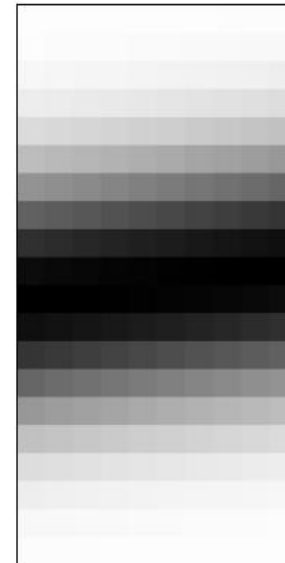
(a) Income



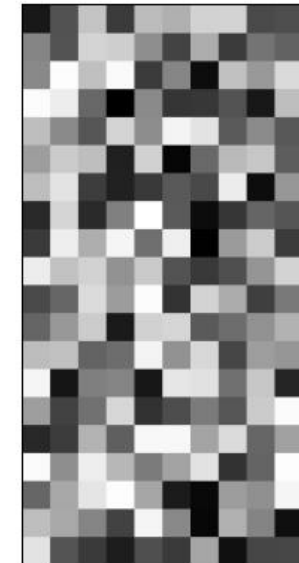
(b) Credit Limit



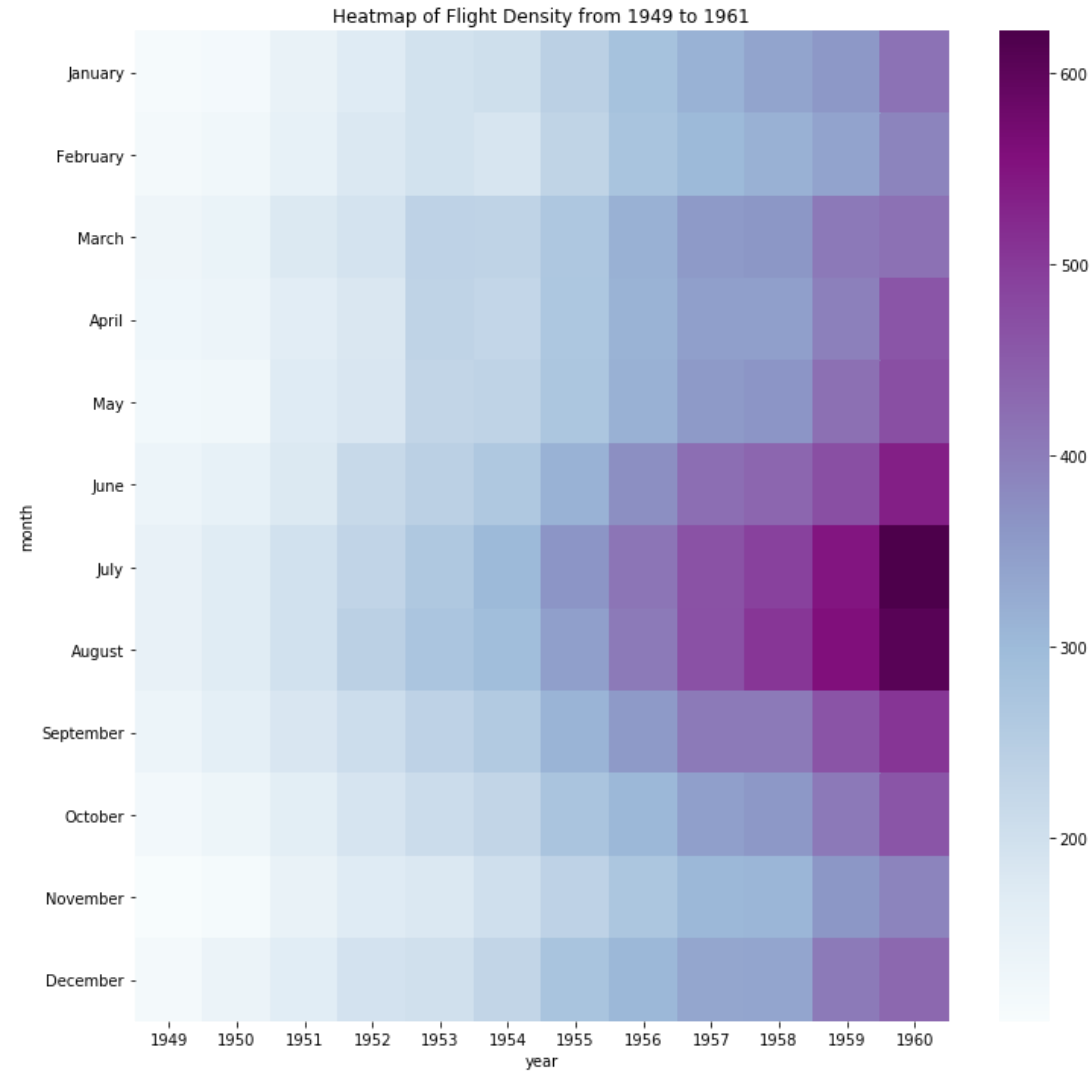
(c) transaction volume



(d) age



Heat Map



Source: <https://blog.algoexhealth.com/2017/09/10-heatmaps-10-python-libraries/>

Geometric Projection

- Scatter plot

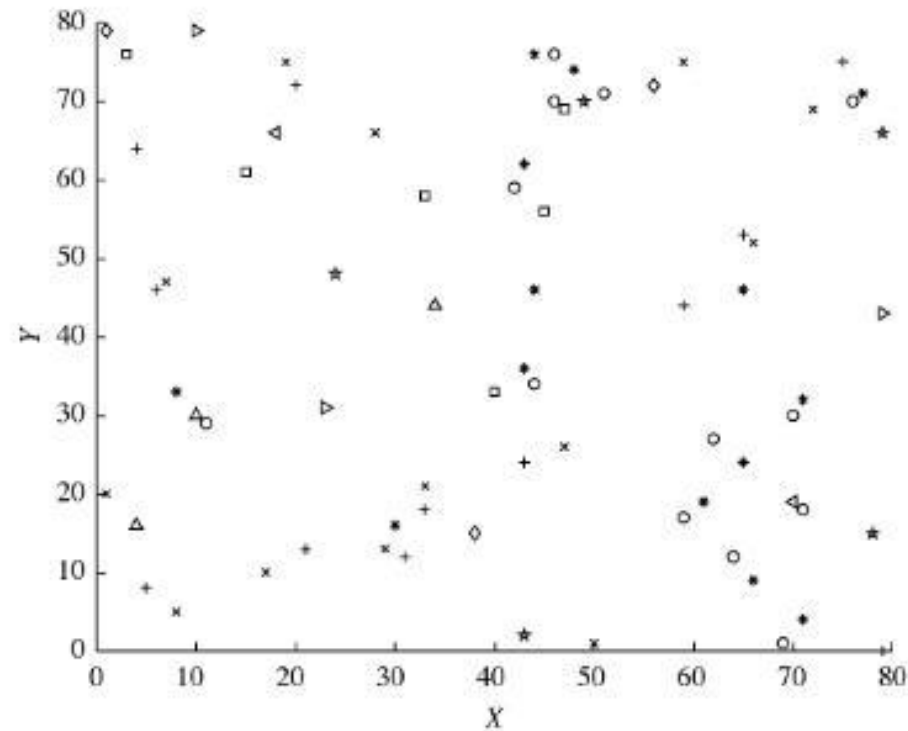
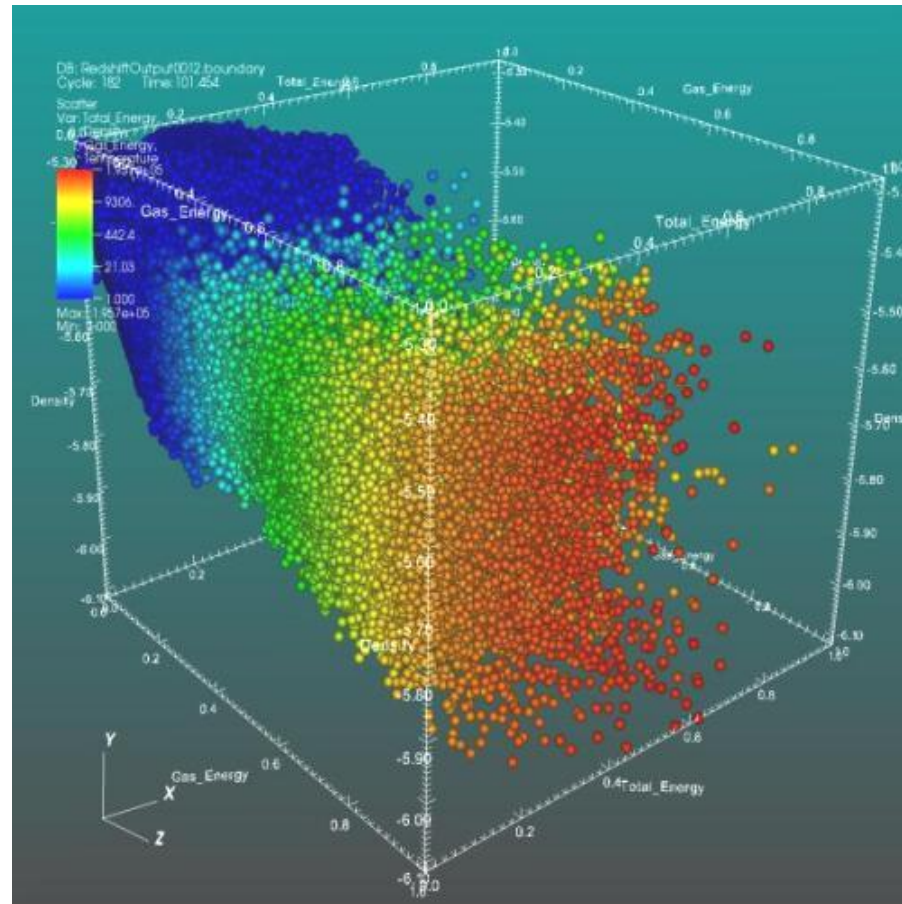


Figure 2.13 Visualization of a 2-D data set using a scatter plot. Source: www.cs.sfu.ca/jpei/publications/rareevent-geoinformatica06.pdf.

Geometric Projection

- 3-D Scatter plot



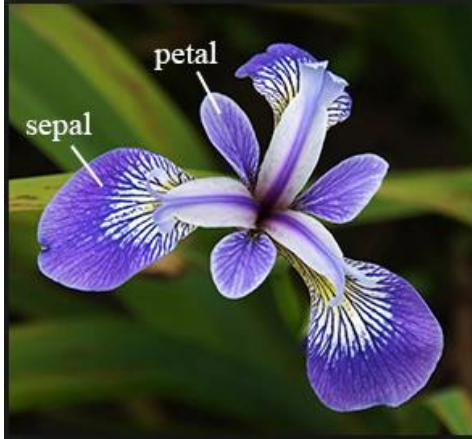
Scatter Plot Matrix

- Iris flower



Source: <https://github.com/Datagatherer2357/Gareth-Duffy-GMIT-Project>

Scatter Plot Matrix

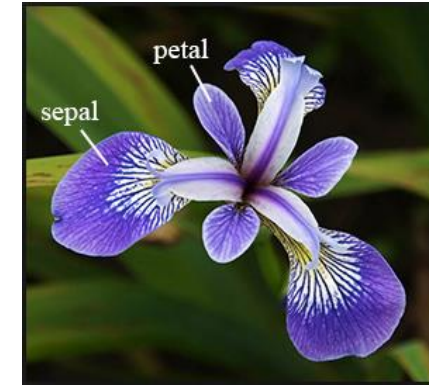
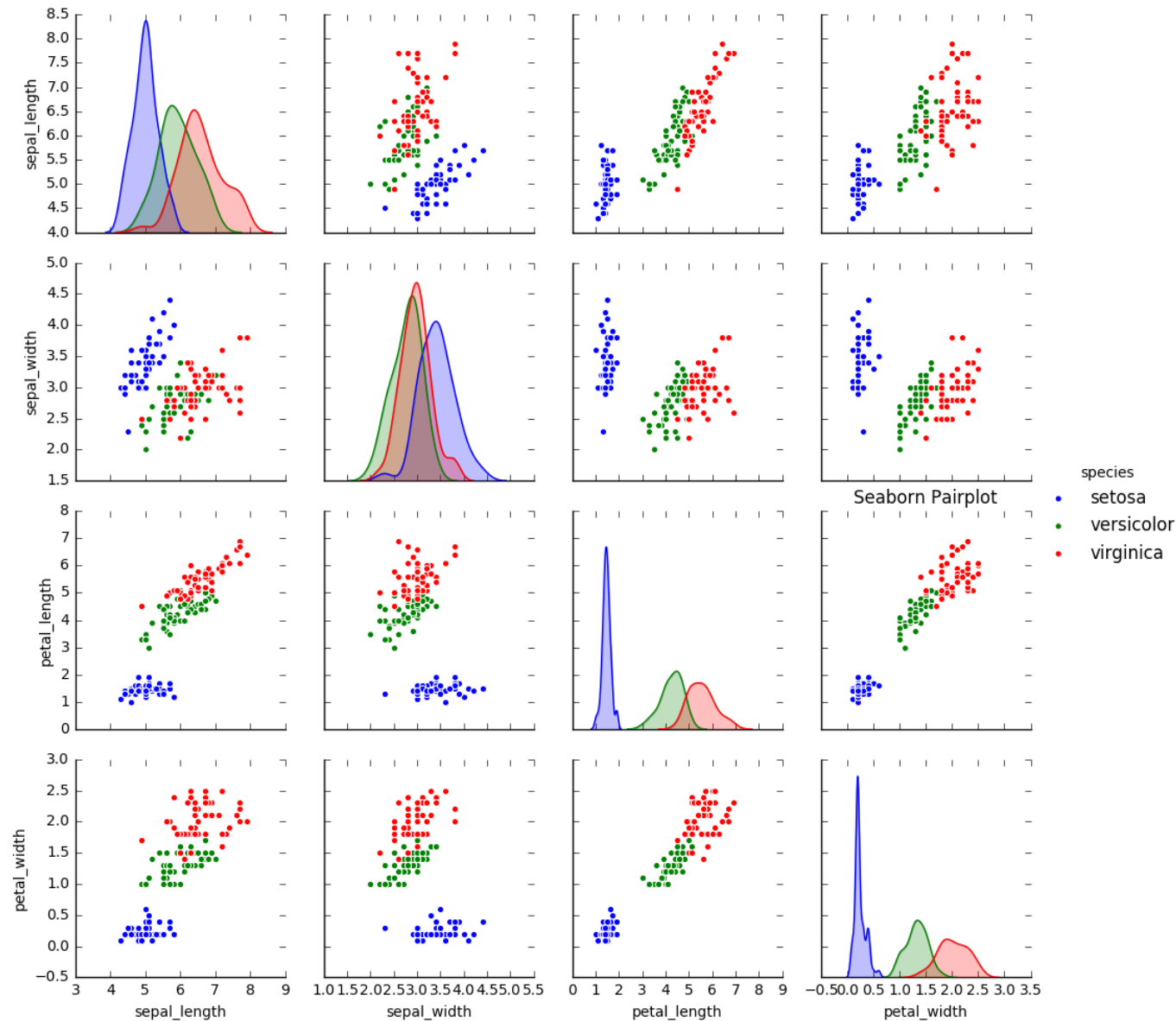


Iris dataset

- 450 samples
- 3 species
- 5 dimensions
 - sepal length
 - sepal width
 - petal length
 - petal width
 - species

	A	B	C	D	E
1	sepal length	sepal width	petal length	petal width	class
2	5.1	3.5	1.4	0.2	Iris-setosa
3	4.9	3	1.4	0.2	Iris-setosa
4	4.7	3.2	1.3	0.2	Iris-setosa
5	4.6	3.1	1.5	0.2	Iris-setosa
6	5	3.6	1.4	0.2	Iris-setosa
7	5.4	3.9	1.7	0.4	Iris-setosa
8	4.6	3.4	1.4	0.3	Iris-setosa
9	5	3.4	1.5	0.2	Iris-setosa
10	4.4	2.9	1.4	0.2	Iris-setosa
11	4.9	3.1	1.5	0.1	Iris-setosa
12	5.4	3.7	1.5	0.2	Iris-setosa
13	4.8	3.4	1.6	0.2	Iris-setosa

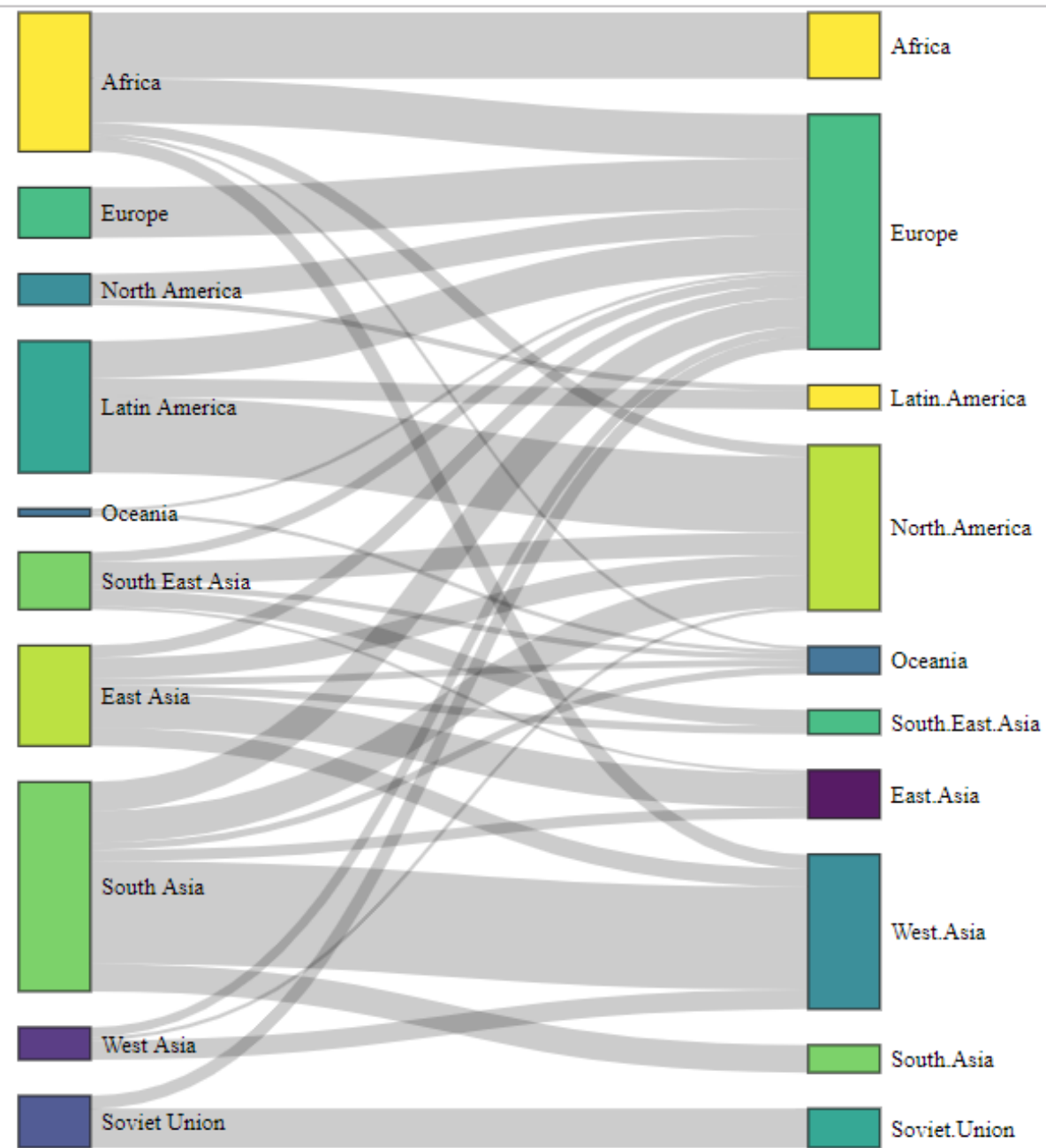
Scatter Plot Matrix



Sankey Diagram

- Sankey Diagram is a visualization technique that allows to display flows.
- Several entities (nodes) are represented by rectangles or text.
- Their links are represented with arrow or arcs that have a width proportional to the importance of the flow.

an example
displaying the
number of people
migrating from
one country (left)
to another (right)



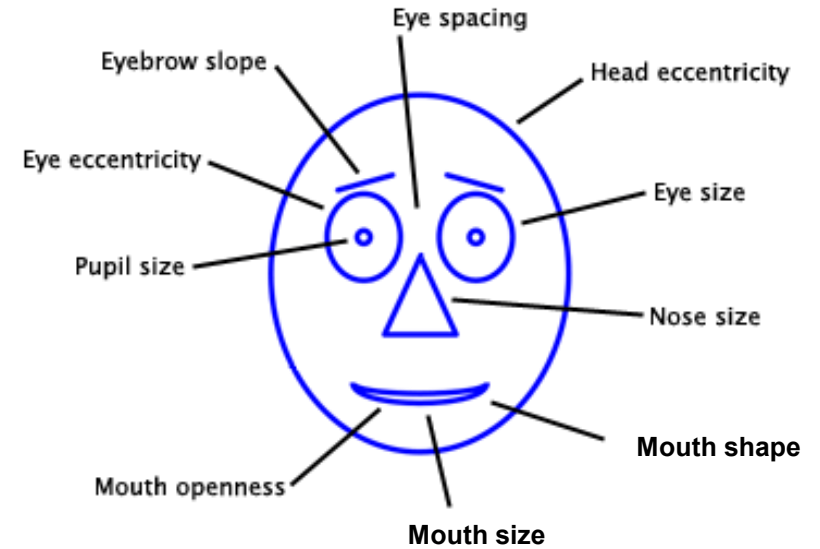
Icon-Based

- Chernoff Faces

- the figure shows faces produced using 10 characteristics:

head eccentricity, eye size, eye spacing, eye eccentricity, pupil size, eyebrow slant, nose size, mouth shape, mouth size, and mouth opening

- each assigned one of 10 possible values, generated using [*Mathematica*](#) (S. Dickson)
- reference: Gonick, L. and Smith, W. [*The Cartoon Guide to Statistics*](#). New York: Harper Perennial, p. 212, 1993

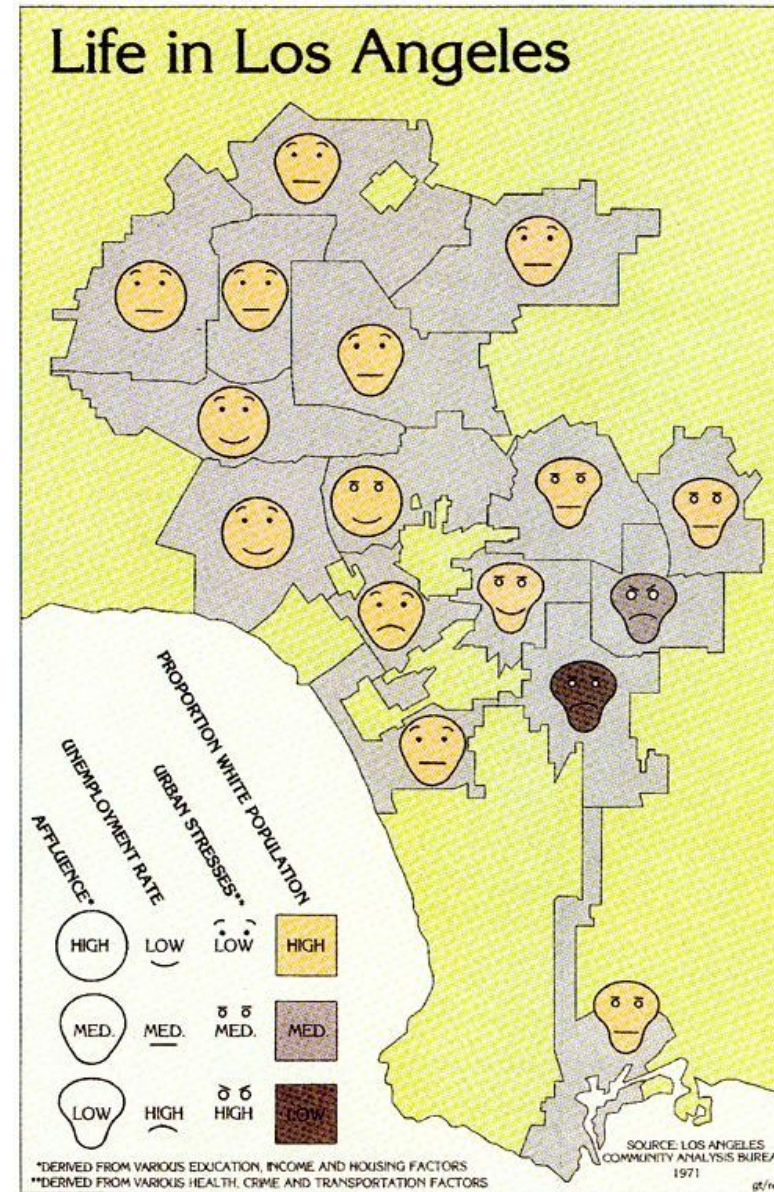


Icon-Based

- Chernoff Faces

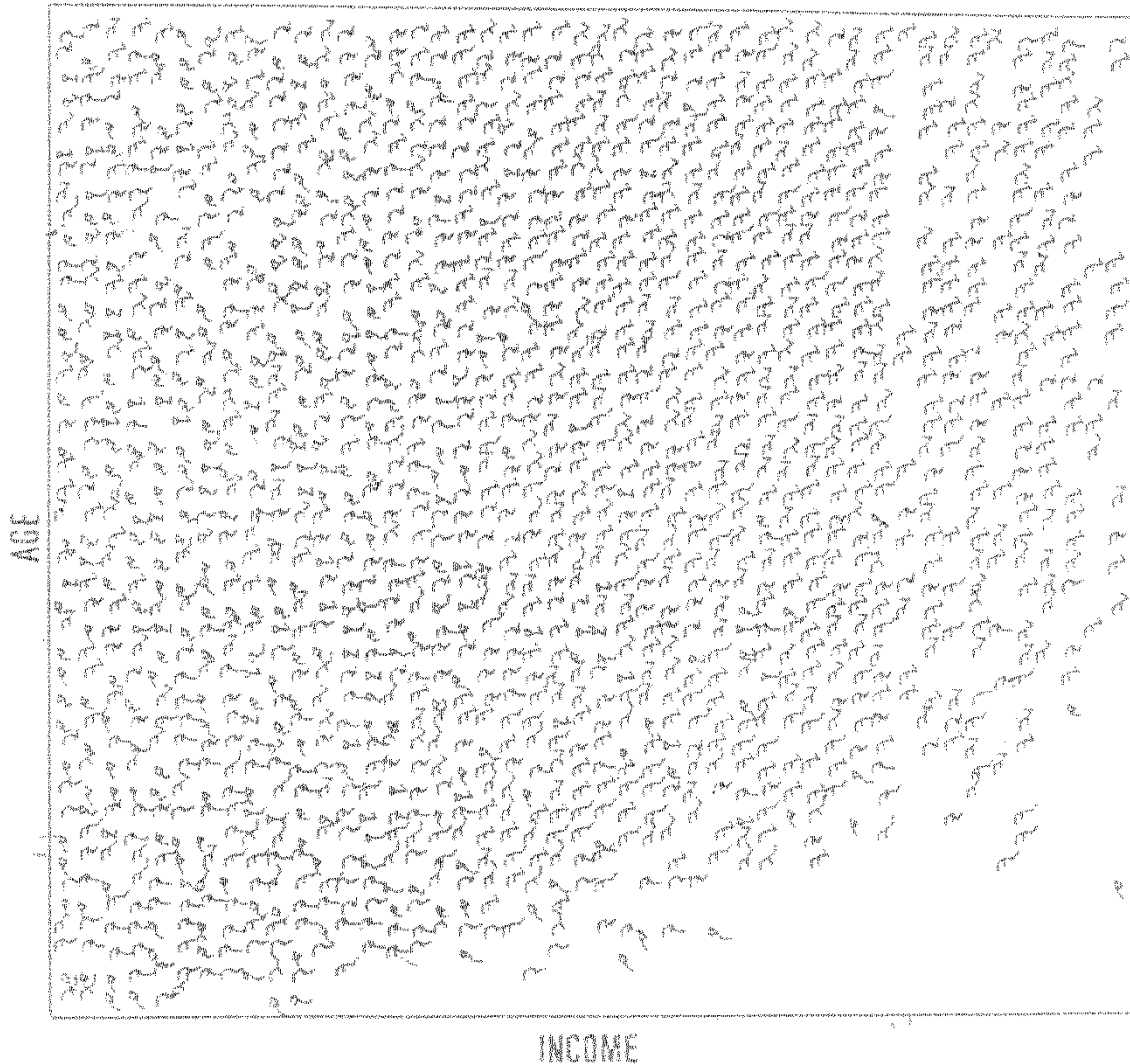
source

<http://kspark.kaist.ac.kr/Human%20Engineering.files/Chernoff/Chernoff%20Faces.htm>



Icon-Based

- Stick Figure

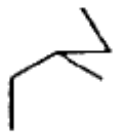
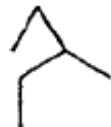
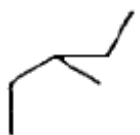
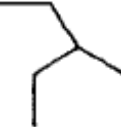
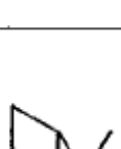
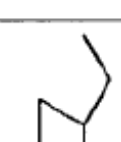
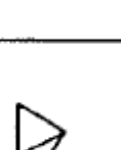
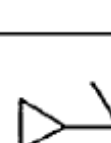
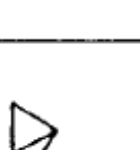


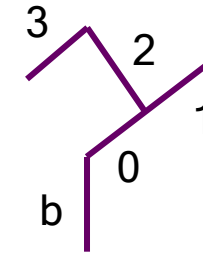
A census data figure showing age, income, gender, education, etc.

A 5-piece stick figure (1 body and 4 limbs w. different angle/length)

Icon-Based

- Stick Figure

	Low Job Married	High Job Married	Low Job Single	High Job Single
Male Low Education				
Male High Education				
Female Low Education				
Female High Education				



Icon Feature	Data Field
b	Reference
0	Sex
1	Education
2	Occupation
3	Marital status

Similarity and Dissimilarity

- **Similarity**
 - Numerical measure of how alike two data objects are
 - Value is higher when objects are more alike
 - Often falls in the range $[0,1]$
- **Dissimilarity** (e.g., distance)
 - Numerical measure of how different two data objects are
 - Lower when objects are more alike
 - Minimum dissimilarity is often 0
 - Upper limit varies

Data Matrix and Dissimilarity Matrix

- Data matrix

- n data points with p dimensions
- Two modes

$$\begin{bmatrix} x_{11} & \dots & x_{1f} & \dots & x_{1p} \\ \dots & \dots & \dots & \dots & \dots \\ x_{i1} & \dots & x_{if} & \dots & x_{ip} \\ \dots & \dots & \dots & \dots & \dots \\ x_{n1} & \dots & x_{nf} & \dots & x_{np} \end{bmatrix}$$

- Dissimilarity matrix

- n data points, but registers only the distance
- A triangular matrix
- Single mode

$$\begin{bmatrix} 0 & & & & \\ d(2,1) & 0 & & & \\ d(3,1) & d(3,2) & 0 & & \\ \vdots & \vdots & \vdots & & \\ d(n,1) & d(n,2) & \dots & \dots & 0 \end{bmatrix}$$

Proximity Measure for Nominal Attributes

- Can take 2 or more states, e.g., red, yellow, blue, green (generalization of a binary attribute)
- Method 1: Simple matching
 - m : # of matches, p : total # of variables

$$d(i, j) = \frac{p - m}{p}$$

- Method 2: Use a large number of binary attributes
 - creating a new binary attribute for each of the M nominal states

Proximity Measure for Binary Attributes

- A contingency table for binary data

		Object j		
		1	0	sum
Object i	1	a	b	$a+b$
	0	c	d	$c+d$
	sum	$a+c$	$b+d$	p

- Distance measure for symmetric binary variables:

$$d(i, j) = \frac{b+c}{a+b+c+d}$$

- Distance measure for asymmetric binary variables:

$$d(i, j) = \frac{b+c}{a+b+c}$$

Dissimilarity between Binary Variables

- Example

Name	Gender	Fever	Cough	Test-1	Test-2	Test-3	Test-4
Jack	M	Y 1	N 0	P 1	N 0	N 0	N 0
Mary	F	Y 1	N 0	P 1	N 0	P 1	N 0
Jim	M	Y 1	P 1	N 0	N 0	N 0	N 0

		Object <i>j</i>		
		1	0	<i>sum</i>
Object <i>i</i>	1	<i>a</i>	<i>b</i>	<i>a+b</i>
	0	<i>c</i>	<i>d</i>	<i>c+d</i>
	<i>sum</i>	<i>a+c</i>	<i>b+d</i>	<i>p</i>

- gender is a symmetric attribute
- the remaining attributes are asymmetric binary
- let the values Y and P be set to 1, and the value N be set to 0

$$d(i, j) = \frac{b + c}{a + b + c}$$

$$d(jack, mary) = \frac{0 + 1}{2 + 0 + 1} = 0.33$$

$$d(jack, jim) = \frac{1 + 1}{1 + 1 + 1} = 0.67$$

$$d(jim, mary) = \frac{1 + 2}{1 + 1 + 2} = 0.75$$

Distance on Numeric Data: Minkowski Distance

- *Minkowski distance*: A popular distance measure

$$d(i, j) = \sqrt[q]{(|x_{i1} - x_{j1}|^q + |x_{i2} - x_{j2}|^q + \dots + |x_{ip} - x_{jp}|^q)}$$

where $i = (x_{i1}, x_{i2}, \dots, x_{ip})$ and $j = (x_{j1}, x_{j2}, \dots, x_{jp})$ are two p -dimensional data objects, and q is the order (the distance so defined is also called L- q norm)

- Properties
 - $d(i, j) > 0$ if $i \neq j$, and $d(i, i) = 0$ (Positive definiteness)
 - $d(i, j) = d(j, i)$ (Symmetry)
 - $d(i, j) \leq d(i, k) + d(k, j)$ (Triangle Inequality)
- A distance that satisfies these properties is a **metric**

Distance on Numeric Data: Minkowski Distance

- $q = 1$: **Manhattan** (city block, L_1 norm) **distance**

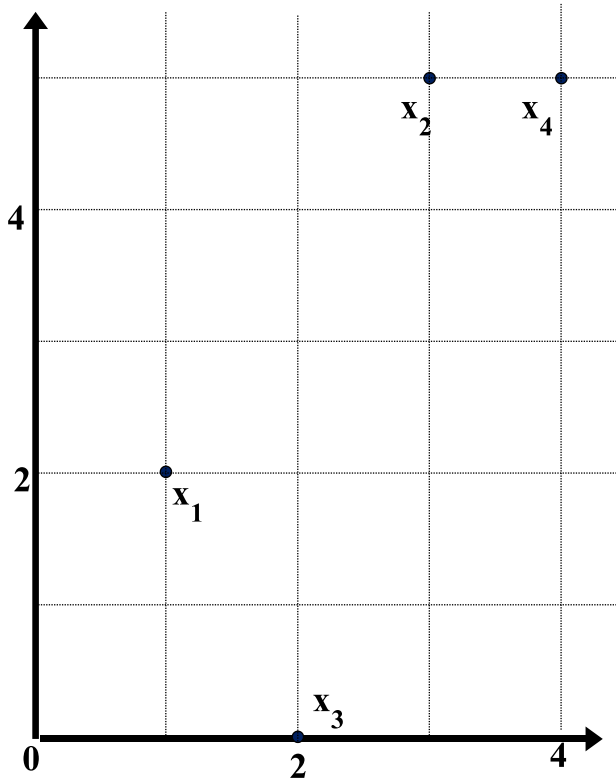
$$d(i, j) = |x_{i_1} - x_{j_1}| + |x_{i_2} - x_{j_2}| + \dots + |x_{i_p} - x_{j_p}|$$

- $q = 2$: (L_2 norm) **Euclidean** distance

$$d(i, j) = \sqrt{(|x_{i_1} - x_{j_1}|^2 + |x_{i_2} - x_{j_2}|^2 + \dots + |x_{i_p} - x_{j_p}|^2)}$$

Example: Minkowski Distance

point	attribute 1	attribute 2
x1	1	2
x2	3	5
x3	2	0
x4	4	5



Dissimilarity Matrices

Manhattan (L_1)

L	x1	x2	x3	x4
x1	0			
x2	5	0		
x3	3	6	0	
x4	6	1	7	0

Euclidean (L_2)

L2	x1	x2	x3	x4
x1	0			
x2	3.61	0		
x3	2.24	5.1	0	
x4	4.24	1	5.39	0

Ordinal Variables

- An ordinal variable can be discrete or continuous
- Order is important, e.g., rank
- Can be treated like interval-scaled
 - replace x_{if} by their rank $r_{if} \in \{1, \dots, M_f\}$
 - map the range of each variable onto $[0, 1]$ by replacing i -th object in the f -th variable by

$$z_{if} = \frac{r_{if} - 1}{M_f - 1}$$

- compute the dissimilarity using methods for interval-scaled variables

Attributes of Mixed Type

- A database may contain all attribute types
 - Nominal, symmetric binary, asymmetric binary, numeric, ordinal
- One may use a weighted formula to combine their effects

$$d(i, j) = \frac{\sum_{f=1}^p \delta_{ij}^{(f)} d_{ij}^{(f)}}{\sum_{f=1}^p \delta_{ij}^{(f)}}$$

- f is binary or nominal:

$d_{ij}^{(f)} = 0$ if $x_{if} = x_{jf}$, or $d_{ij}^{(f)} = 1$ otherwise

- f is numeric: use the normalized distance

$$d(i, j) = \frac{|x_{if} - x_{jf}|}{\max - \min}$$

- f is ordinal

- Compute ranks r_{if} and
- Treat z_{if} as interval-scaled

$$z_{if} = \frac{r_{if} - 1}{M_f - 1}$$

Attributes of Mixed Type

where $\delta_{ij}^{(f)} = 0$ if either

- (1) x_{if} or x_{jf} is missing (i.e., there is no measurement of attribute f for object i or object j), or
- (2) $x_{if} = x_{jf} = 0$ and attribute f is asymmetric binary;

$\delta_{ij}^{(f)} = 1$ otherwise,

Example: Attributes of Mixed Type

Object ID	Test-1 (nominal)	Test-2 (ordinal)	Test-3 (numeric)
1	code-A	excellent	45
2	code-B	fair	22
3	code-C	good	64
4	code-A	excellent	28

Nominal: Test-1

$$d(i, j) = \begin{bmatrix} 0 & & & \\ 1 & 0 & & \\ 1 & 1 & 0 & \\ 0 & 1 & 1 & 0 \end{bmatrix}$$

Example: Attributes of Mixed Type

Object ID	Test-1 (nominal)	Test-2 (ordinal)	Test-3 (numeric)
1	code-A	3 excellent 1	45
2	code-B	1 fair 0	22
3	code-C	2 good 0.5	64
4	code-A	3 excellent 1	28

Ordinal: Test-2

1. Replace x_{if} with r_{if}
2. normalize r_{if} in range $[0,1]$
3. Compute Euclidean distance

$$z_{if} = \frac{r_{if} - 1}{M_f - 1}$$

$$\text{if } r_{if} = 1 \quad z_{if} = \frac{1 - 1}{3 - 1} = 0$$

$$\text{if } r_{if} = 2 \quad z_{if} = \frac{2 - 1}{3 - 1} = 0.5$$

$$\text{if } r_{if} = 3 \quad z_{if} = \frac{3 - 1}{3 - 1} = 1$$

$$d(2,1) = \sqrt{(0 - 1)^2}$$

$$d(i,j) = \begin{bmatrix} 0 & & & \\ 1 & 0 & & \\ 0.5 & 0.5 & 0 & \\ 0 & 1 & 0.5 & 0 \end{bmatrix}$$

Example: Attributes of Mixed Type

Object ID	Test-1 (nominal)	Test-2 (ordinal)	Test-3 (numeric)
1	code-A	excellent	45
2	code-B	fair	22
3	code-C	good	64
4	code-A	excellent	28

Numeric: Test-3

normalized distance

$$d(i,j) = \frac{|x_{if} - x_{jf}|}{\max - \min}$$

$$d(2,1) = \frac{|22 - 45|}{64 - 22} = \frac{23}{42} = 0.55 \quad d(i,j) = \begin{bmatrix} 0 & & & \\ 0.55 & 0 & & \\ 0.45 & 1 & 0 & \\ 0.40 & 0.14 & 0.86 & 0 \end{bmatrix}$$

Example: Attributes of Mixed Type

Mixed type:

Test-1

$$d(i,j) = \begin{bmatrix} 0 & & & \\ \textcircled{1} & 0 & & \\ 1 & 1 & 0 & \\ 0 & 1 & 1 & 0 \end{bmatrix}$$

Test-2

$$d(i,j) = \begin{bmatrix} 0 & & & \\ \textcircled{1} & 0 & & \\ 0.5 & 0.5 & 0 & \\ 0 & 1 & 0.5 & 0 \end{bmatrix}$$

Test-3

$$d(i,j) = \begin{bmatrix} 0 & & & \\ \textcircled{0.55} & 0 & & \\ 0.45 & 1 & 0 & \\ 0.40 & 0.14 & 0.86 & 0 \end{bmatrix}$$

$$d(i,j) = \frac{\sum_{f=1}^p \delta_{ij}^{(f)} d_{ij}^{(f)}}{\sum_{f=1}^p \delta_{ij}^{(f)}}$$

$$d(2,1) = \frac{(1 \times \textcircled{1}) + (1 \times \textcircled{1}) + (1 \times \textcircled{0.55})}{1 + 1 + 1} = 0.85$$

$$d(i,j) = \begin{bmatrix} 0 & & & \\ 0.85 & 0 & & \\ 0.65 & 0.83 & 0 & \\ 0.13 & 0.71 & 0.79 & 0 \end{bmatrix}$$

Cosine Similarity

- A **document** can be represented by thousands of attributes, each recording the *frequency* of a particular word (such as keywords) or phrase in the document.

<i>Document</i>	<i>teamcoach</i>		<i>hockey</i>	<i>baseball</i>	<i>soccer</i>	<i>penalty</i>	<i>score</i>	<i>win</i>	<i>loss</i>	<i>season</i>
Document1	5	0	3	0	2	0	0	2	0	0
Document2	3	0	2	0	1	1	0	1	0	1
Document3	0	7	0	2	1	0	0	3	0	0
Document4	0	1	0	0	1	2	2	0	3	0

- Cosine measure: If d_1 and d_2 are two vectors (e.g., term-frequency vectors), then
$$\cos(d_1, d_2) = (d_1 \bullet d_2) / \|d_1\| \|d_2\| ,$$
where $\bullet$ indicates vector dot product,
 $\|d\|$: the length of vector d

Example: Cosine Similarity

- $\cos(d_1, d_2) = (d_1 \bullet d_2) / \|d_1\| \|d_2\|$,
where $\bullet$ indicates vector dot product, $\|d\|$: the length of vector d

- Ex: Find the **similarity** between documents 1 and 2.

$$d_1 = (5, 0, 3, 0, 2, 0, 0, 2, 0, 0)$$

$$d_2 = (3, 0, 2, 0, 1, 1, 0, 1, 0, 1)$$

$$d_1 \bullet d_2 = 5*3 + 0*0 + 3*2 + 0*0 + 2*1 + 0*1 + 0*1 + 2*1 + 0*0 + 0*1 = 25$$

$$\|d_1\| = (5*5 + 0*0 + 3*3 + 0*0 + 2*2 + 0*0 + 0*0 + 2*2 + 0*0 + 0*0)^{0.5} = (42)^{0.5} = 6.481$$

$$\|d_2\| = (3*3 + 0*0 + 2*2 + 0*0 + 1*1 + 1*1 + 0*0 + 1*1 + 0*0 + 1*1)^{0.5} = (17)^{0.5} = 4.12$$

$$\cos(d_1, d_2) = 0.94$$